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Section: 5B

Lab 10

Task 1 — GitHub CLI Codespace Setup & Authentication

Goal: Install GH CLI (if not present), authenticate for Codespace use, and connect into the Codespace shell (all steps inside Codespace unless noted).

Install GitHub CLI:

winget install --id GitHub.cli

```
C:\Users\HABIB TARIQ>winget install --id GitHub.cli
Found an existing package already installed. Trying to upgrade the installed package...
No available upgrade found.
No newer package versions are available from the configured sources.
```

Authenticate GH CLI for Codespaces:

gh auth login -s codespace

When prompted, generate a GitHub access token (classic). Set scopes to: admin:org, codespace, repo.

```
C:\Users\HABIB TARIQ>gh auth login -s codespace
? Where do you use GitHub? GitHub.com
? What is your preferred protocol for Git operations on this host? HTTPS
? Authenticate Git with your GitHub credentials? Yes
? How would you like to authenticate GitHub CLI? Login with a web browser

! First copy your one-time code: 312F-BAB9
Press Enter to open https://github.com/login/device in your browser...
✓ Authentication complete.
- gh config set -h github.com git_protocol https
✓ Configured git protocol
✓ Logged in as manahilhabib031
! You were already logged in to this account
```

List available codespaces:

gh codespace list

```
C:\Users\HABIB TARIQ>gh codespace list
NAME          DISPLAY NAME  REPOSITORY  BRANCH  STATE    CREATED AT
expert-space... expert sp...  manahilha... main*    Shutdown about 9 d...

C:\Users\HABIB TARIQ>|
```

Connect to a codespace via SSH:

gh codespace ssh -c <name_of_codespace>

```
C:\Users\HABIB TARIQ>gh codespace list
NAME          DISPLAY NAME  REPOSITORY  BRANCH  STATE    CREATED AT
expert-space... expert sp...  manahilha... main*    Shutdown about 9 d...

C:\Users\HABIB TARIQ>gh codespace ssh -c expert-space-pancake-pj5r5959vvrxf
qqw
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
Last login: Fri Jan  9 02:54:49 2026 from ::1
@manahilhabib031 → /workspaces/Lab9 (main) $ |
```

Task 2 — Install AWS CLI, Terraform CLI, Provider Setup

Goal: Install AWS CLI (if not present) & Terraform CLI inside Codespace; configure initial Terraform provider.

A. Install AWS CLI (Skip if already installed)

In Codespace shell, install AWS CLI:

```
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
```

```
unzip awscliv2.zip
```

```
sudo ./aws/install
```

```
aws --version
```

```
@manahilhabib031 → /workspaces/Lab9 (main) $ aws --version
aws-cli/2.32.26 Python/3.13.11 Linux/6.8.0-1030-azure exe/x86_64.ubuntu.24
@manahilhabib031 → /workspaces/Lab9 (main) $ |
```

Configure AWS CLI:

```
aws configure
```

```
@manahilhabib031 → /workspaces/Lab9 (main) $ aws configure
AWS Access Key ID [*****IJUV]: AKIARMIC7HAZCXVXIJUV
AWS Secret Access Key [*****5hMW]: qQB0RxHL36a1aXGk
Pz65ds05hMW
Default region name [me-central-1]:
Default output format [json]:
@manahilhabib031 → /workspaces/Lab9 (main) $ |
```

Then check config files:

```
cat ~/.aws/credentials
```

```
cat ~/.aws/config
```

```
@manahilhabib031 → /workspaces/Lab9 (main) $ cat ~/.aws/credentials
[default]
aws_access_key_id = AKIARMIC7HAZB6NRVCOG
aws_secret_access_key = 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
@manahilhabib031 → /workspaces/Lab9 (main) $ cat ~/.aws/config
[default]
region = me-central-1
output = json
```

Verify AWS CLI connectivity:

```
aws sts get-caller-identity
```

```
@manahilhabib031 → /workspaces/Lab9 (main) $ aws sts get-caller-identity
{
  "UserId": "AIDARMIC7HAZDK6HLORCS",
  "Account": "095032391730",
  "Arn": "arn:aws:iam::095032391730:user/labs"
}
```

B. Install Terraform CLI

Install Terraform CLI:

```
wget -O - https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o
/usr/share/keyrings/hashicorp-archive-keyring.gpg
```

```
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/hashicorp-archive-
keyring.gpg] https://apt.releases.hashicorp.com $(grep -oP '(?<=UBUNTU_CODENAME=).*'
/etc/os-release || lsb_release -cs) main" | sudo tee /etc/apt/sources.list.d/hashicorp.list
```

```
sudo apt update
```

```
sudo apt install terraform
```

```
which terraform
```

```
terraform --version
```

```

@manahilhabib031 → /workspaces/Lab9 (main) $ sudo apt install terraform
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  terraform
0 upgraded, 1 newly installed, 0 to remove and 22 not upgraded.
Need to get 30.6 MB of archives.
After this operation, 101 MB of additional disk space will be used.
Get:1 https://apt.releases.hashicorp.com noble/main amd64 terraform amd64 1
.14.3-1 [30.6 MB]
Fetched 30.6 MB in 0s (75.3 MB/s)
Selecting previously unselected package terraform.
(Reading database ... 58657 files and directories currently installed.)
Preparing to unpack .../terraform_1.14.3-1_amd64.deb ...
Unpacking terraform (1.14.3-1) ...
Setting up terraform (1.14.3-1) ...
@manahilhabib031 → /workspaces/Lab9 (main) $ terraform --version
Terraform v1.14.3
on linux_amd64
@manahilhabib031 → /workspaces/Lab9 (main) $ |

```

C. Provider Configuration (main.tf)

Create main.tf using Vim:

vim main.tf

```

@manahilhabib031 → /workspaces/Lab9 (main) $ vim main.tf
@manahilhabib031 → /workspaces/Lab9 (main) $ |

```

```

provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

```

```

provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

```

Initialize Terraform:

terraform init

```

@manahilhabib031 → /workspaces/Lab9 (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.28.0...
- Installed hashicorp/aws v6.28.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provide
r
selections it made above. Include this file in your version control reposit
ory
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to s
ee
any changes that are required for your infrastructure. All Terraform comman
ds
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, o
ther
commands will detect it and remind you to do so if necessary.

```

Show contents of .terraform.lock.hcl:

cat .terraform.lock.hcl

```

@manahilhabib031 → /workspaces/Lab9 (main) $ cat .terraform.lock.hcl
# This file is maintained automatically by "terraform init".
# Manual edits may be lost in future updates.

provider "registry.terraform.io/hashicorp/aws" {
  version = "6.28.0"
  hashes = [
    "h1:wzZdGs0FFmNqIgPyo9tKnGKJ37BGNSgwRrEXayL29+0=",
    "zh:0ba0d5eb6e0c6a933eb2bef3c3dbf22b58fbc0337bf138f95bf0e8bb6e6df93e",
    "zh:23eacd4e6db32cf0ff2ce189461bdbb62e46513978d33c5de4decc4670870ec",
    "zh:307b06a15fc00a8e6fd243abde2cbe5112e9d40371542665b91bec1018dd6e3c",
    "zh:37a02d5b45a9d050b9642c9e2e268297254192280df72f6e46641daca52e40ec",
    "zh:3da866639f07d92e734557d673092719c33ede80f4276c835bf7f231a669aa33",
    "zh:480060b0ba310d0f6b6a14d60b276698cb103c48fd2f7e2802ae47c963995ec6",
    "zh:57796453455c20db80d9168edbf125bf6180e1aae869de1546a2be58e4e405ec",
    "zh:69139cba772d4df8de87598d8d8a2b1b4b254866db046c061dccc79edb14e6b9",
    "zh:7312763259b859ff911c5452ca8bdf7d0be6231c5ea0de2df8f09d51770900ac",
    "zh:8d2d6f4015d3c155d7eb53e36f019a729aefb46ebfe13f3a637327d3a1402ecc",
    "zh:94ce589275c77308e6253f607de96919b840c2dd36c44aa798f693c9dd81af42",
    "zh:9b12af85486a96aedd8d7984b0ff811a4b42e3d88dad1a3fb4c0b580d04fa425",
    "zh:adaceec6a1bf4f5df1e12bd72cf52b72087c72efed078aef636f8988325b1a8b",
    "zh:d37be1ce187d94fd9df7b13a717c219964cd835c946243f096c6b230cdfd7e92",
    "zh:fe6205b5ca2ff36e68395cb8d3ae10a3728f405cdbc46b206a515e1ebcf17a1",
  ]
}
@manahilhabib031 → /workspaces/Lab9 (main) $ |

```

Show contents of .terraform/ directory:

ls .terraform/

```
@manahilhabib031 → /workspaces/Lab9 (main) $ ls .terraform/
providers
@manahilhabib031 → /workspaces/Lab9 (main) $ |
```

Task 3 — VPC/Subnet Creation, Initialization, Verification

Goal: Use Terraform to create VPC & Subnet resources, verify via CLI.

Edit main.tf to add:

```
resource "aws_vpc" "development_vpc" {
  cidr_block = "10.0.0.0/16"
}

resource "aws_subnet" "dev_subnet_1" {
  vpc_id     = aws_vpc.development_vpc.id
  cidr_block = "10.0.10.0/24"
  availability_zone = "me-central-1a"
}
```

Use vim main.tf, edit and insert code.

```
provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "development_vpc" {
  cidr_block = "10.0.0.0/16"
}

resource "aws_subnet" "dev_subnet_1" {
  vpc_id           = aws_vpc.development_vpc.id
  cidr_block       = "10.0.10.0/24"
  availability_zone = "me-central-1a"
}
```

Run:

terraform apply

Type yes when prompted.

```
@manahilhabib031 → /workspaces/Lab9 (main) $ terraform apply

Terraform used the selected providers to generate the following execution
plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1 will be created
+ resource "aws_subnet" "dev_subnet_1" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                          = "10.0.10.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                  = (known after apply)
  + ipv6_cidr_block_association_id      = (known after apply)
  + ipv6_native                         = false
  + map_public_ip_on_launch             = false
  + owner_id                            = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                              = "me-central-1"
  + tags_all                            = (known after apply)
  + vpc_id                              = (known after apply)
}
```

```
# aws_vpc.development_vpc will be created
+ resource "aws_vpc" "development_vpc" {
  + arn                                = (known after apply)
  + cidr_block                          = "10.0.0.0/16"
  + default_network_acl_id              = (known after apply)
  + default_route_table_id              = (known after apply)
  + default_security_group_id           = (known after apply)
  + dhcp_options_id                     = (known after apply)
  + enable_dns_hostnames                 = (known after apply)
  + enable_dns_support                   = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                                  = (known after apply)
  + instance_tenancy                     = "default"
  + ipv6_association_id                  = (known after apply)
  + ipv6_cidr_block                      = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id                  = (known after apply)
  + owner_id                             = (known after apply)
  + region                              = "me-central-1"
  + tags_all                             = (known after apply)
}
```

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_vpc.development_vpc: Creating...

aws_vpc.development_vpc: Creation complete after 2s [id=vpc-08dd77cd2066b8855]

aws_subnet.dev_subnet_1: Creating...

aws_subnet.dev_subnet_1: Creation complete after 1s [id=subnet-0bb5a852509006120]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

@manahilhabib031 → /workspaces/Lab9 (main) \$

Verify resources using AWS CLI:

Run:

```
aws ec2 describe-subnets --filter "Name=subnet-id,Values=<subnet-id>"
```

```
@manahilhabib031 → /workspaces/Lab9 (main) $ aws ec2 describe-subnets \
--filters "Name=subnet-id,Values=subnet-0f284e40a191a3e62"
{
  "Subnets": [
    {
      "AvailabilityZoneId": "mec1-az3",
      "MapCustomerOwnedIpOnLaunch": false,
      "OwnerId": "095032391730",
      "AssignIpv6AddressOnCreation": false,
      "Ipv6CidrBlockAssociationSet": [],
      "SubnetArn": "arn:aws:ec2:me-central-1:095032391730:subnet/subn
et-0f284e40a191a3e62",
      "EnableDns64": false,
      "Ipv6Native": false,
      "PrivateDnsNameOptionsOnLaunch": {
        "HostnameType": "ip-name",
        "EnableResourceNameDnsARecord": false,
        "EnableResourceNameDnsAAAARecord": false
      },
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "SubnetId": "subnet-0f284e40a191a3e62",
      "State": "available",
      "VpcId": "vpc-021a5e129ca204b3f",
      "CidrBlock": "172.31.0.0/20",
      "AvailableIpAddressCount": 4090,
      "AvailabilityZone": "me-central-1c",
      "DefaultForAz": true,
      "MapPublicIpOnLaunch": true
    }
  ]
}
```

Run:

```
aws ec2 describe-vpcs --filter "Name=vpc-id,Values=<vpc-id>"
```



```
@manahilhabib031 → /workspaces/Lab9 (main) $ aws ec2 describe-vpcs \
--filters "Name=vpc-id,Values=vpc-021a5e129ca204b3f"
{
  "Vpcs": [
    {
      "OwnerId": "095032391730",
      "InstanceTenancy": "default",
      "CidrBlockAssociationSet": [
        {
          "AssociationId": "vpc-cidr-assoc-0236b7082da231dc3",
          "CidrBlock": "172.31.0.0/16",
          "CidrBlockState": {
            "State": "associated"
          }
        }
      ],
      "IsDefault": true,
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "VpcId": "vpc-021a5e129ca204b3f",
      "State": "available",
      "CidrBlock": "172.31.0.0/16",
      "DhcpOptionsId": "dopt-06b319f01537b2a69"
    }
  ]
}
@manahilhabib031 → /workspaces/Lab9 (main) $ |
```

Task 4 — Data Source, Targeted Destroy, Tags

A. Data Source & Resource Creation

Add to main.tf:

```
data "aws_vpc" "existing_vpc" {
  default = true
}

resource "aws_subnet" "dev_subnet_1_existing" {
  vpc_id    = data.aws_vpc.existing_vpc.id
  cidr_block = "172.31.48.0/24"
  availability_zone = "me-central-1a"
}
```

Use vim to edit and insert code.

```

provider "aws" {
  shared_config_files      = ["~/aws/config"]
  shared_credentials_files = ["~/aws/credentials"]
}
resource "aws_vpc" "development_vpc" {
  cidr_block = "10.0.0.0/16"
}

resource "aws_subnet" "dev_subnet_1" {
  vpc_id            = aws_vpc.development_vpc.id
  cidr_block        = "10.0.10.0/24"
  availability_zone  = "me-central-1a"
}
data "aws_vpc" "existing_vpc" {
  default = true
}

resource "aws_subnet" "dev_subnet_1_existing" {
  vpc_id            = data.aws_vpc.existing_vpc.id
  cidr_block        = "172.31.48.0/24"
  availability_zone  = "me-central-1a"
}

```

Apply configuration:

terraform apply

Confirm only the new subnet is created.

```

@manahilhabib031 → /workspaces/Lab9 (main) $ terraform apply
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-08dd77cd2066b8855]
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-021a5e129ca204b3f]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0bb5a852509006120]

Terraform used the selected providers to generate the following execution
plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id               = (known after apply)
  + cidr_block                        = "172.31.48.0/24"
  + enable_dns64                      = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch            = false
  + owner_id                          = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                            = "me-central-1"
  + tags_all                          = (known after apply)
  + vpc_id                            = "vpc-021a5e129ca204b3f"
}

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_subnet.dev_subnet_1_existing: Creating...
aws_subnet.dev_subnet_1_existing: Creation complete after 2s [id=subnet-0362f37a799b91c28]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
@manahilhabib031 → /workspaces/Lab9 (main) $ |

```

B. Targeted Destroy & Refresh

Destroy only one resource:

terraform destroy -target=aws_subnet.dev_subnet_1_existing

```
@manahilhabib031 → /workspaces/Lab9 (main) $ terraform destroy -target=aws_subnet.dev_subnet_1_existing
data.aws_vpc.existing_vpc: Reading...
data.aws_vpc.existing_vpc: Read complete after 2s [id=vpc-021a5e129ca204b3f]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-0362f37a799b91c28]

Terraform used the selected providers to generate the following execution plan.
Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1_existing will be destroyed
- resource "aws_subnet" "dev_subnet_1_existing" {
  - arn                                = "arn:aws:ec2:me-central-1:095032391730:subnet/subnet-0362f37a799b91c28" -> null
  - assign_ipv6_address_on_creation    = false -> null
  - availability_zone                  = "me-central-1a" -> null
  - availability_zone_id                = "mec1-az1" -> null
  - cidr_block                         = "172.31.48.0/24" -> null
  - enable_dns64                       = false -> null
  - enable_lni_at_device_index         = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
  - id                                 = "subnet-0362f37a799b91c28" -> null
  - ipv6_native                        = false -> null
  - map_customer_owned_ip_on_launch    = false -> null
  - map_public_ip_on_launch            = false -> null
  - owner_id                           = "095032391730" -> null
  - private_dns_hostname_type_on_launch = "ip-name" -> null
  - region                             = "me-central-1" -> null
  - tags                               = {} -> null
  - tags_all                           = {} -> null
  - vpc_id                             = "vpc-021a5e129ca204b3f" -> null
  # (4 unchanged attributes hidden)
}

Plan: 0 to add, 0 to change, 1 to destroy.

Warning: Resource targeting is in effect

You are creating a plan with the -target option, which means that the result of
this plan may not represent all of the changes requested by the current
configuration.

The -target option is not for routine use, and is provided only for exceptional
situations such as recovering from errors or mistakes, or when Terraform
specifically suggests to use it as part of an error message.

Do you really want to destroy all resources?
Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

aws_subnet.dev_subnet_1_existing: Destroying... [id=subnet-0362f37a799b91c28]
aws_subnet.dev_subnet_1_existing: Destruction complete after 1s

Warning: Applied changes may be incomplete

The plan was created with the -target option in effect, so some changes
requested in the configuration may have been ignored and the output values may
not be fully updated. Run the following command to verify that no other changes
are pending:
    terraform plan

Note that the -target option is not suitable for routine use, and is provided
only for exceptional situations such as recovering from errors or mistakes, or
when Terraform specifically suggests to use it as part of an error message.

Destroy complete! Resources: 1 destroyed.
@manahilhabib031 → /workspaces/Lab9 (main) $ |
```

Refresh state:

terraform refresh

```
@manahilhabib031 → /workspaces/Lab9 (main) $ terraform refresh
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-08dd77cd2066b8855]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0bb5a852509006120]
data.aws_vpc.existing_vpc: Read complete after 2s [id=vpc-021a5e129ca204b3f]
@manahilhabib031 → /workspaces/Lab9 (main) $
```

Re-apply configuration:

terraform apply

```
@manahilhabib031 → /workspaces/Lab9 (main) $ terraform apply
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-08dd77cd2066b8855]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0bb5a852509006120]
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-021a5e129ca204b3f]

Terraform used the selected providers to generate the following execution
plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                               = (known after apply)
  + assign_ipv6_address_on_creation   = false
  + availability_zone                 = "me-central-1a"
  + availability_zone_id              = (known after apply)
  + cidr_block                        = "172.31.48.0/24"
  + enable_dns64                      = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                = (known after apply)
  + ipv6_cidr_block_association_id    = (known after apply)
  + ipv6_native                       = false
  + map_public_ip_on_launch           = false
  + owner_id                          = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                            = "me-central-1"
  + tags_all                          = (known after apply)
  + vpc_id                            = "vpc-021a5e129ca204b3f"
}

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_subnet.dev_subnet_1_existing: Creating...
aws_subnet.dev_subnet_1_existing: Creation complete after 0s [id=subnet-01f73f1afc8016232]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
@manahilhabib031 → /workspaces/Lab9 (main) $
```

Destroy all resources:

terraform destroy

```
@manahilhabib031 → /workspaces/Lab9 (main) $ terraform destroy
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-08dd77cd2066b8855]
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-021a5e129ca204b3f]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-01f73f1afc8016232]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0bb5a852509006120]

Terraform used the selected providers to generate the following execution plan.
Resource actions are indicated with the following symbols:
  - destroy

Terraform will perform the following actions:

  # aws_subnet.dev_subnet_1 will be destroyed
  - resource "aws_subnet" "dev_subnet_1" {
    - arn                                = "arn:aws:ec2:me-central-1:095032391730:subnet/subnet-0bb5a852509006120" -> null
    - assign_ipv6_address_on_creation    = false -> null
    - availability_zone                  = "me-central-1a" -> null
    - availability_zone_id                = "mec1-az1" -> null
    - cidr_block                         = "10.0.10.0/24" -> null
    - enable_dns64                       = false -> null
    - enable_lni_at_device_index         = 0 -> null
    - enable_resource_name_dns_a_record_on_launch = false -> null
    - enable_resource_name_dns_aaaa_record_on_launch = false -> null
    - id                                 = "subnet-0bb5a852509006120" -> null
    - ipv6_native                        = false -> null
    - map_customer_owned_ip_on_launch    = false -> null
    - map_public_ip_on_launch            = false -> null
    - owner_id                           = "095032391730" -> null
    - private_dns_hostname_type_on_launch = "ip-name" -> null
    - region                             = "me-central-1" -> null
    - tags                               = {} -> null
    - tags_all                           = {} -> null
    - vpc_id                             = "vpc-08dd77cd2066b8855" -> null
    # (4 unchanged attributes hidden)
  }

  # aws_subnet.dev_subnet_1_existing will be destroyed
  - resource "aws_subnet" "dev_subnet_1_existing" {
    - arn                                = "arn:aws:ec2:me-central-1:095032391730:subnet/subnet-01f73f1afc8016232" -> null
    - assign_ipv6_address_on_creation    = false -> null
    - availability_zone                  = "me-central-1a" -> null
    - availability_zone_id                = "mec1-az1" -> null
    - cidr_block                         = "10.0.10.0/24" -> null
    - enable_dns64                       = false -> null
    - enable_lni_at_device_index         = 0 -> null
    - enable_resource_name_dns_a_record_on_launch = false -> null
    - enable_resource_name_dns_aaaa_record_on_launch = false -> null
    - id                                 = "subnet-01f73f1afc8016232" -> null
    - ipv6_native                        = false -> null
    - map_customer_owned_ip_on_launch    = false -> null
    - map_public_ip_on_launch            = false -> null
    - owner_id                           = "095032391730" -> null
    - private_dns_hostname_type_on_launch = "ip-name" -> null
    - region                             = "me-central-1" -> null
    - tags                               = {} -> null
    - tags_all                           = {} -> null
    - vpc_id                             = "vpc-08dd77cd2066b8855" -> null
    # (4 unchanged attributes hidden)
  }

Plan: 0 to add, 0 to change, 3 to destroy.

Do you really want to destroy all resources?
Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

aws_subnet.dev_subnet_1_existing: Destroying... [id=subnet-01f73f1afc8016232]
aws_subnet.dev_subnet_1: Destroying... [id=subnet-0bb5a852509006120]
aws_subnet.dev_subnet_1: Destruction complete after 0s
aws_vpc.development_vpc: Destroying... [id=vpc-08dd77cd2066b8855]
aws_subnet.dev_subnet_1_existing: Destruction complete after 1s
aws_vpc.development_vpc: Destruction complete after 0s

Destroy complete! Resources: 3 destroyed.
@manahilhabib031 → /workspaces/Lab9 (main) $ |
```

Plan configuration:

terraform plan

```
@manahilhabib031 → /workspaces/Lab9 (main) $ terraform plan
data.aws_vpc.existing_vpc: Reading...
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-021a5e129ca204b3f]

Terraform used the selected providers to generate the following execution plan. Resource actions
are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1 will be created
+ resource "aws_subnet" "dev_subnet_1" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "10.0.10.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                         = false
  + map_public_ip_on_launch            = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
  + tags_all                           = (known after apply)
  + vpc_id                             = (known after apply)
}

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "172.31.48.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                         = false
}
```

```
# aws_vpc.development_vpc will be created
+ resource "aws_vpc" "development_vpc" {
  + arn                                = (known after apply)
  + cidr_block                        = "10.0.0.0/16"
  + default_network_acl_id           = (known after apply)
  + default_route_table_id           = (known after apply)
  + default_security_group_id        = (known after apply)
  + dhcp_options_id                  = (known after apply)
  + enable_dns_hostnames              = (known after apply)
  + enable_dns_support               = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                               = (known after apply)
  + instance_tenancy                  = "default"
  + ipv6_association_id              = (known after apply)
  + ipv6_cidr_block                  = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id              = (known after apply)
  + owner_id                         = (known after apply)
  + region                           = "me-central-1"
  + tags_all                         = (known after apply)
}
```

Plan: 3 to add, 0 to change, 0 to destroy.

Note: You didn't use the `-out` option to save this plan, so Terraform can't guarantee to take exactly these actions if you run `"terraform apply"` now.

@manahilhabib031 → /workspaces/Lab9 (main) \$ |

Apply again:

terraform apply

```

@manahilhabib031 → /workspaces/Lab9 (main) $ terraform apply
data.aws_vpc.existing_vpc: Reading...
data.aws_vpc.existing_vpc: Read complete after 2s [id=vpc-021a5e129ca204b3f]

Terraform used the selected providers to generate the following execution plan. Resource actions
are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1 will be created
+ resource "aws_subnet" "dev_subnet_1" {
+   arn                                = (known after apply)
+   assign_ipv6_address_on_creation    = false
+   availability_zone                  = "me-central-1a"
+   availability_zone_id                = (known after apply)
+   cidr_block                         = "10.0.10.0/24"
+   enable_dns64                       = false
+   enable_resource_name_dns_a_record_on_launch = false
+   enable_resource_name_dns_aaaa_record_on_launch = false
+   id                                 = (known after apply)
+   ipv6_cidr_block_association_id     = (known after apply)
+   ipv6_native                        = false
+   map_public_ip_on_launch            = false
+   owner_id                           = (known after apply)
+   private_dns_hostname_type_on_launch = (known after apply)
+   region                             = "me-central-1"
+   tags_all                           = (known after apply)
+   vpc_id                             = (known after apply)
}

# aws_subnet.dev_subnet_1_existing will be created
+ resource "aws_subnet" "dev_subnet_1_existing" {
+   arn                                = (known after apply)
+   assign_ipv6_address_on_creation    = false
+   availability_zone                  = "me-central-1a"
+   availability_zone_id                = (known after apply)
+   cidr_block                         = "172.31.48.0/24"
+   enable_dns64                       = false
+   enable_resource_name_dns_a_record_on_launch = false
+   enable_resource_name_dns_aaaa_record_on_launch = false
+   id                                 = (known after apply)
+   ipv6_cidr_block_association_id     = (known after apply)
}

# aws_vpc.development_vpc will be created
+ resource "aws_vpc" "development_vpc" {
+   arn                                = (known after apply)
+   cidr_block                         = "10.0.0.0/16"
+   default_network_acl_id             = (known after apply)
+   default_route_table_id             = (known after apply)
+   default_security_group_id          = (known after apply)
+   dhcp_options_id                   = (known after apply)
+   enable_dns_hostnames               = (known after apply)
+   enable_dns_support                 = true
+   enable_network_address_usage_metrics = (known after apply)
+   id                                 = (known after apply)
+   instance_tenancy                   = "default"
+   ipv6_association_id                = (known after apply)
+   ipv6_cidr_block                    = (known after apply)
+   ipv6_cidr_block_network_border_group = (known after apply)
+   main_route_table_id                = (known after apply)
+   owner_id                           = (known after apply)
+   region                             = "me-central-1"
+   tags_all                           = (known after apply)
}

Plan: 3 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

Enter a value: yes

aws_subnet.dev_subnet_1_existing: Creating...
aws_vpc.development_vpc: Creating...
aws_subnet.dev_subnet_1_existing: Creation complete after 0s [id=subnet-058fc8aced9ba1f0f]
aws_vpc.development_vpc: Creation complete after 2s [id=vpc-0fe303f2e8e85c238]
aws_subnet.dev_subnet_1: Creating...
aws_subnet.dev_subnet_1: Creation complete after 1s [id=subnet-0f29654ba6145927f]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
@manahilhabib031 → /workspaces/Lab9 (main) $

```

C. Tagging Resources

Modify main.tf to add tags:

```
resource "aws_vpc" "development_vpc"
  cidr_block = "10.0.0.0/16"
  tags = {
    Name: "development"
    vpc_env = "dev"
  }
}

resource "aws_subnet" "dev_subnet_1" {
  vpc_id    = aws_vpc.development_vpc.id
  cidr_block = "10.0.10.0/24"
  availability_zone = "me-central-1a"
  tags = {
    Name: "subnet-1-dev"
  }
}

resource "aws_subnet" "dev_subnet_1_existing" {
  vpc_id    = data.aws_vpc.existing_vpc.id
  cidr_block = "172.31.48.0/24"
  availability_zone = "me-central-1a"
  tags = {
    Name: "subnet-1-default"
  }
}
```

Use vim to update tags.

```

provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "development_vpc" {
  cidr_block = "10.0.0.0/16"

  tags = {
    Name     = "development"
    vpc_env = "dev"
  }
}

resource "aws_subnet" "dev_subnet_1" {
  vpc_id            = aws_vpc.development_vpc.id
  cidr_block        = "10.0.10.0/24"
  availability_zone = "me-central-1a"

  tags = {
    Name = "subnet-1-dev"
  }
}

resource "aws_subnet" "dev_subnet_1_existing" {
  vpc_id            = data.aws_vpc.existing_vpc.id
  cidr_block        = "172.31.48.0/24"
  availability_zone = "me-central-1a"

  tags = {
    Name = "subnet-1-default"
  }
}

```

Run:

terraform refresh

terraform apply -auto-approve

```

@manahilhabib031 → /workspaces/Lab9 (main) $ terraform refresh
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-0fe303f2e8e85c238]
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-021a5e129ca204b3f]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-058fc8aced9ba1f0f]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0f29654ba6145927f]
@manahilhabib031 → /workspaces/Lab9 (main) $ |

```

```

@manahilhabib031 → /workspaces/Lab9 (main) $ terraform apply -auto-approve
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-0fe303f2e8e85c238]
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-021a5e129ca204b3f]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-058fc8aced9ba1f0f]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0f29654ba6145927f]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated
with the following symbols:
  ~ update in-place

Terraform will perform the following actions:

# aws_vpc.development_vpc will be updated in-place
~ resource "aws_vpc" "development_vpc" {
  id          = "vpc-0fe303f2e8e85c238"
  ~ tags      = {
    "Name" = "development"
    - "vpc_env" = "dev" -> null
  }
  ~ tags_all  = {
    - "vpc_env" = "dev" -> null
    # (1 unchanged element hidden)
  }
  # (19 unchanged attributes hidden)
}

Plan: 0 to add, 1 to change, 0 to destroy.
aws_vpc.development_vpc: Modifying... [id=vpc-0fe303f2e8e85c238]
aws_vpc.development_vpc: Modifications complete after 1s [id=vpc-0fe303f2e8e85c238]

Apply complete! Resources: 0 added, 1 changed, 0 destroyed.
@manahilhabib031 → /workspaces/Lab9 (main) $ |

```

Remove vpc_env = "dev" tag from development_vpc resource, re-plan/apply:

```

@manahilhabib031 → /workspaces/Lab9 (main) $ terraform plan
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-0fe303f2e8e85c238]
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-021a5e129ca204b3f]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-058fc8aced9ba1f0f]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0f29654ba6145927f]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated
with the following symbols:
  ~ update in-place

Terraform will perform the following actions:

# aws_vpc.development_vpc will be updated in-place
~ resource "aws_vpc" "development_vpc" {
  id          = "vpc-0fe303f2e8e85c238"
  ~ tags      = {
    "Name" = "development"
    - "vpc_env" = "dev" -> null
  }
  ~ tags_all  = {
    - "vpc_env" = "dev" -> null
    # (1 unchanged element hidden)
  }
  # (19 unchanged attributes hidden)
}

Plan: 0 to add, 1 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these
actions if you run "terraform apply" now.
@manahilhabib031 → /workspaces/Lab9 (main) $ |

```

```

ca. Command Prompt - gh codespace ssh
provider "aws" {
  shared_config_files = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

# Only 1 VPC
resource "aws_vpc" "development_vpc" {
  cidr_block = "10.0.0.0/16"
  tags = {
    Name = "development"
  }
}

# Only 1 dev subnet in this VPC
resource "aws_subnet" "dev_subnet_1" {
  vpc_id = aws_vpc.development_vpc.id
  cidr_block = "10.0.10.0/24"
  availability_zone = "me-central-1a"
  tags = {
    Name = "subnet-1-dev"
  }
}

# Data source for existing VPC
data "aws_vpc" "existing_vpc" {
  default = true
}

# Only 1 subnet in the existing VPC
resource "aws_subnet" "dev_subnet_1_existing" {
  vpc_id = data.aws_vpc.existing_vpc.id
  cidr_block = "172.31.48.0/24"
  availability_zone = "me-central-1a"
  tags = {
    Name = "subnet-1-default"
  }
}

```

```

@manahilhabib031 → /workspaces/Lab9 (main) $ terraform plan
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-0fe303f2e8e85c238]
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-021a5e129ca204b3f]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-058fc8aced9ba1f0f]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0f29654ba6145927f]

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- ~ update in-place

Terraform will perform the following actions:

```

# aws_vpc.development_vpc will be updated in-place
~ resource "aws_vpc" "development_vpc" {
  id = "vpc-0fe303f2e8e85c238"
  ~ tags = {
    "Name" = "development"
    - "vpc_env" = "dev" -> null
  }
  ~ tags_all = {
    - "vpc_env" = "dev" -> null
    # (1 unchanged element hidden)
  }
  # (19 unchanged attributes hidden)
}

```

Plan: 0 to add, 1 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

```

@manahilhabib031 → /workspaces/Lab9 (main) $ |

```

```

@manahilhabib031 → /workspaces/Lab9 (main) $ terraform apply -
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-0fe303f2e
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-021a
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subn
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0f2965

Terraform used the selected providers to generate the following
with the following symbols:
  ~ update in-place

Terraform will perform the following actions:

# aws_vpc.development_vpc will be updated in-place
~ resource "aws_vpc" "development_vpc" {
    id                = "vpc-0fe303f2e8e85c23"
    ~ tags            = {
        "Name"      = "development"
        - "vpc_env" = "dev" -> null
    }
    ~ tags_all        = {
        - "vpc_env" = "dev" -> null
        # (1 unchanged element hidden)
    }
    # (19 unchanged attributes hidden)
}

Plan: 0 to add, 1 to change, 0 to destroy.
aws_vpc.development_vpc: Modifying... [id=vpc-0fe303f2e8e85c23]
aws_vpc.development_vpc: Modifications complete after 1s [id=v

```

Task 5 — State File Inspection & Terraform State Commands

Destroy all resources:

terraform destroy

```

@manahilhabib031 → /workspaces/Lab9 (main) $ terraform destroy
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-0fe303f2e8e85c238]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0f29654ba6145927f]
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-021a5e129ca204b3f]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-058fc8aced9balf0f]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated
with the following symbols:
  - destroy

Terraform will perform the following actions:

# aws_subnet.dev_subnet_1 will be destroyed
- resource "aws_subnet" "dev_subnet_1" {
  - arn                                = "arn:aws:ec2:me-central-1:095032391730:subnet/subnet-0f2
9654ba6145927f" -> null
  - assign_ipv6_address_on_creation    = false -> null
  - availability_zone                  = "me-central-1a" -> null
  - availability_zone_id                = "mec1-az1" -> null
  - cidr_block                         = "10.0.10.0/24" -> null
  - enable_dns64                       = false -> null
  - enable_lni_at_device_index         = 0 -> null
  - enable_resource_name_dns_a_record_on_launch = false -> null
  - enable_resource_name_dns_aaaa_record_on_launch = false -> null
  - id                                 = "subnet-0f29654ba6145927f" -> null
  - ipv6_native                        = false -> null
  - map_customer_owned_ip_on_launch    = false -> null
  - map_public_ip_on_launch            = false -> null
  - owner_id                           = "095032391730" -> null
  - private_dns_hostname_type_on_launch = "ip-name" -> null
  - region                             = "me-central-1" -> null
  - tags                               = {
    - "Name" = "subnet-1-dev"
  } -> null
  - tags_all                           = {
    - "Name" = "subnet-1-dev"
  } -> null
  - vpc_id                             = "vpc-0fe303f2e8e85c238" -> null
  # (4 unchanged attributes hidden)
}

# aws_subnet.dev_subnet_1_existing will be destroyed
- resource "aws_subnet" "dev_subnet_1_existing" {
  - arn                                = "arn:aws:ec2:me-central-1:095032391730:subnet/subnet-058
fc8aced9balf0f" -> null
  - assign_ipv6_address_on_creation    = false -> null
  - availability_zone                  = "me-central-1a" -> null

```

```
# aws_vpc.development_vpc will be destroyed
- resource "aws_vpc" "development_vpc" {
  - arn = "arn:aws:ec2:me-central-1:784
  - assign_generated_ipv6_cidr_block = false -> null
  - cidr_block = "10.0.0.0/16" -> null
  - default_network_acl_id = "acl-08a7bb4f5c6f7e605" -> nu
  - default_route_table_id = "rtb-071ad852849d25444" -> nu
  - default_security_group_id = "sg-0009f7fe9168e487d" -> nul
  - dhcp_options_id = "dopt-073394897c6005777" -> n
  - enable_dns_hostnames = false -> null
  - enable_dns_support = true -> null
  - enable_network_address_usage_metrics = false -> null
  - id = "vpc-0c15bd9a6c886ba2b" -> nu
  - instance_tenancy = "default" -> null
  - ipv6_netmask_length = 0 -> null
  - main_route_table_id = "rtb-071ad852849d25444" -> nu
  - owner_id = "784386063361" -> null
  - region = "me-central-1" -> null
  - tags = {
    - "Name" = "development"
  } -> null
  - tags_all = {
    - "Name" = "development"
  } -> null
  # (4 unchanged attributes hidden)
}
```

Plan: 0 to add, 0 to change, 3 to destroy.

Do you really want to destroy all resources?

Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

```
aws_subnet.dev_subnet_1_existing: Destroying... [id=subnet-04010169e79a73202]
aws_subnet.dev_subnet_1: Destroying... [id=subnet-0b3abd0b4ad1170d4]
aws_subnet.dev_subnet_1_existing: Destruction complete after 1s
aws_subnet.dev_subnet_1: Destruction complete after 1s
aws_vpc.development_vpc: Destroying... [id=vpc-0c15bd9a6c886ba2b]
aws_vpc.development_vpc: Destruction complete after 0s
```

Inspect state files after destroy:

Run:

```
cat terraform.tfstate
```

```
Destroy complete! Resources: 3 destroyed.  
@manahilhabib031 → /workspaces/Lab9 (main) $ cat terraform.tfstate  
{  
  "version": 4,  
  "terraform_version": "1.14.3",  
  "serial": 31,  
  "lineage": "8b1d5d5f-e9bc-aaf3-332c-e079db2b49e2",  
  "outputs": {},  
  "resources": [],  
  "check_results": null  
}  
@manahilhabib031 → /workspaces/Lab9 (main) $ |
```

Run:

cat terraform.tfstate.backup


```

@manahilhabib031 → /workspaces/Lab9 (main) $ cat terraform.tfstate.backup
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 26,
  "lineage": "8b1d5d5f-e9bc-aaf3-332c-e079db2b49e2",
  "outputs": {},
  "resources": [
    {
      "mode": "data",
      "type": "aws_vpc",
      "name": "existing_vpc",
      "provider": "provider[\"registry.terraform.io/hashicorp/aws\"]",
      "instances": [
        {
          "schema_version": 0,
          "attributes": {
            "arn": "arn:aws:ec2:me-central-1:095032391730:vpc/vpc-021a5e129ca204b3f",
            "cidr_block": "172.31.0.0/16",
            "cidr_block_associations": [
              {
                "association_id": "vpc-cidr-assoc-0236b7082da231dc3",
                "cidr_block": "172.31.0.0/16",
                "state": "associated"
              }
            ],
            "default": true,
            "dhcp_options_id": "dopt-06b319f01537b2a69",
            "enable_dns_hostnames": true,
            "enable_dns_support": true,
            "enable_network_address_usage_metrics": false,
            "filter": null,
            "id": "vpc-021a5e129ca204b3f",
            "instance_tenancy": "default",
            "ipv6_association_id": "",
            "ipv6_cidr_block": "",
            "main_route_table_id": "rtb-01d7dcdb9b4e75165",
            "owner_id": "095032391730",
            "region": "me-central-1",
            "state": null,
            "tags": {},
            "timeouts": null
          },
          "sensitive_attributes": [],
          "identity_schema_version": 0
        }
      ]
    },
    {
      "mode": "managed",

```

Recreate resources:

terraform apply

```
@manahilhabib031 → /workspaces/Lab9 (main) $ terraform apply
```

```
data.aws_vpc.existing_vpc: Reading...
```

```
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-021a5e129ca204b3f]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

aws_subnet.dev_subnet_1 will be created

```
+ resource "aws_subnet" "dev_subnet_1" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "10.0.10.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id      = (known after apply)
  + ipv6_native                         = false
  + map_public_ip_on_launch            = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
  + tags                               = {
    + "Name" = "subnet-1-dev"
  }
  + tags_all                           = {
    + "Name" = "subnet-1-dev"
  }
  + vpc_id                             = (known after apply)
}
```

aws_subnet.dev_subnet_1_existing will be created

```
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "172.31.48.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id      = (known after apply)
  + ipv6_native                         = false
  + map_public_ip_on_launch            = false
  + owner_id                           = (known after apply)
}
```

```

+ resource "aws_vpc" "development_vpc" {
  + arn                    = (known after apply)
  + cidr_block             = "10.0.0.0/16"
  + default_network_acl_id = (known after apply)
  + default_route_table_id = (known after apply)
  + default_security_group_id = (known after apply)
  + dhcp_options_id        = (known after apply)
  + enable_dns_hostnames    = (known after apply)
  + enable_dns_support      = true
  + enable_network_address_usage_metrics = (known after apply)
  + id                     = (known after apply)
  + instance_tenancy        = "default"
  + ipv6_association_id     = (known after apply)
  + ipv6_cidr_block         = (known after apply)
  + ipv6_cidr_block_network_border_group = (known after apply)
  + main_route_table_id     = (known after apply)
  + owner_id                = (known after apply)
  + region                  = "me-central-1"
  + tags                    = {
    + "Name" = "development"
  }
  + tags_all                = {
    + "Name" = "development"
  }
}

```

Plan: 3 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

aws_subnet.dev_subnet_1_existing: Creating...

aws_vpc.development_vpc: Creating...

aws_subnet.dev_subnet_1_existing: Creation complete after 1s [id=subnet-06e543931482a05cc]

aws_vpc.development_vpc: Creation complete after 2s [id=vpc-0ff8ba45dd1117b74]

aws_subnet.dev_subnet_1: Creating...

aws_subnet.dev_subnet_1: Creation complete after 1s [id=subnet-08d3345f7fcf2eb31]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

View state files:

Run:

cat terraform.tfstate

```

@manahilhabib031 → /workspaces/Lab9 (main) $ cat terraform.tfstate
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 35,
  "lineage": "8b1d5d5f-e9bc-aaf3-332c-e079db2b49e2",
  "outputs": {},
  "resources": [
    {
      "mode": "data",
      "type": "aws_vpc",
      "name": "existing_vpc",
      "provider": "provider[\"registry.terraform.io/hashicorp/aws\"]",
      "instances": [
        {
          "schema_version": 0,
          "attributes": {
            "arn": "arn:aws:ec2:me-central-1:095032391730:vpc/vpc-021a5e129ca204b3f",
            "cidr_block": "172.31.0.0/16",
            "cidr_block_associations": [
              {
                "association_id": "vpc-cidr-assoc-0236b7082da231dc3",
                "cidr_block": "172.31.0.0/16",
                "state": "associated"
              }
            ],
            "default": true,
            "dhcp_options_id": "dopt-06b319f01537b2a69",
            "enable_dns_hostnames": true,
            "enable_dns_support": true,
            "enable_network_address_usage_metrics": false,
            "filter": null,
            "id": "vpc-021a5e129ca204b3f",
            "instance_tenancy": "default",
            "ipv6_association_id": "",
            "ipv6_cidr_block": "",
            "main_route_table_id": "rtb-01d7dcdb9b4e75165",
            "owner_id": "095032391730",
            "region": "me-central-1",
            "state": null,
            "tags": {},
            "timeouts": null
          },
          "sensitive_attributes": [],
          "identity_schema_version": 0
        }
      ],
      "mode": "managed",
      "type": "aws_subnet",
      "name": "dev_subnet_1",
      "provider": "provider[\"registry.terraform.io/hashicorp/aws\"]",

```

```

    "assign_generated_ipv6_cidr_block": false,
    "cidr_block": "10.0.0.0/16",
    "default_network_acl_id": "acl-05c081e1808f62e56",
    "default_route_table_id": "rtb-019157c5b7f37a0fa",
    "default_security_group_id": "sg-027d74c38c1dea210",
    "dhcp_options_id": "dopt-073394897c6005777",
    "enable_dns_hostnames": false,
    "enable_dns_support": true,
    "enable_network_address_usage_metrics": false,
    "id": "vpc-0ff8ba45dd1117b74",
    "instance_tenancy": "default",
    "ipv4_ipam_pool_id": null,
    "ipv4_netmask_length": null,
    "ipv6_association_id": "",
    "ipv6_cidr_block": "",
    "ipv6_cidr_block_network_border_group": "",
    "ipv6_ipam_pool_id": "",
    "ipv6_netmask_length": 0,
    "main_route_table_id": "rtb-019157c5b7f37a0fa",
    "owner_id": "784386063361",
    "region": "me-central-1",
    "tags": {
      "Name": "development"
    },
    "tags_all": {
      "Name": "development"
    }
  },
  "sensitive_attributes": [],
  "identity_schema_version": 0,
  "identity": {
    "account_id": "784386063361",
    "id": "vpc-0ff8ba45dd1117b74",
    "region": "me-central-1"
  },
  "private": "eyJzY2h1bWFFdmVyc2lvbiI6IjEifQ=="
}
]
}
],
"check_results": null

```

Run:

cat terraform.tfstate.backup

```

@manahilhabib031 → /workspaces/Lab9 (main) $ cat terraform.tfstate.backup
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 31,
  "lineage": "8b1d5d5f-e9bc-aaf3-332c-e079db2b49e2",
  "outputs": {},
  "resources": [],
  "check_results": null
}
@manahilhabib031 → /workspaces/Lab9 (main) $ |

```

List resources:

terraform state list

```
@manahilhabib031 → /workspaces/Lab9 (main) $ terraform state list
data.aws_vpc.existing_vpc
aws_subnet.dev_subnet_1
aws_subnet.dev_subnet_1_existing
aws_vpc.development_vpc
@manahilhabib031 → /workspaces/Lab9 (main) $ |
```

Show full attributes:

terraform state show <resource-name>

```
@manahilhabib031 → /workspaces/Lab9 (main) $ terraform state show aws_vpc.development_vpc
# aws_vpc.development_vpc:
resource "aws_vpc" "development_vpc" {
  arn                               = "arn:aws:ec2:me-central-1:095032391730:vpc/vpc-08964a93a7cb32cac"
  assign_generated_ipv6_cidr_block  = false
  cidr_block                        = "10.0.0.0/16"
  default_network_acl_id           = "acl-0809fe7d56d251919"
  default_route_table_id           = "rtb-0b74e2c9e367c6b18"
  default_security_group_id        = "sg-0110838c664125c11"
  dhcp_options_id                  = "dopt-06b319f01537b2a69"
  enable_dns_hostnames             = false
  enable_dns_support               = true
  enable_network_address_usage_metrics = false
  id                               = "vpc-08964a93a7cb32cac"
  instance_tenancy                 = "default"
  ipv6_association_id              = null
  ipv6_cidr_block                  = null
  ipv6_cidr_block_network_border_group = null
  ipv6_ipam_pool_id               = null
  ipv6_netmask_length              = 0
  main_route_table_id              = "rtb-0b74e2c9e367c6b18"
  owner_id                        = "095032391730"
  region                           = "me-central-1"
  tags                             = {
    "Name" = "development"
  }
  tags_all                         = {
    "Name" = "development"
  }
}
```



Task 6 — Terraform Outputs & Attributes Reporting

Add outputs in main.tf:

```
output "dev-vpc-id" {
  value = aws_vpc.development_vpc.id
}

output "dev-subnet-id" {
  value = aws_subnet.dev_subnet_1.id
}

output "dev-vpc-arn" {
```

```

    value = aws_vpc.development_vpc.arn
  }
  output "dev-subnet-arn" {
    value = aws_subnet.dev_subnet_1.arn
  }

```

```

# Only 1 subnet in the existing VPC
resource "aws_subnet" "dev_subnet_1_existing" {
  vpc_id            = data.aws_vpc.existing_vpc.id
  cidr_block        = "172.31.48.0/24"
  availability_zone = "me-central-1a"
  tags = {
    Name = "subnet-1-default"
  }
}
output "dev-vpc-id" {
  value = aws_vpc.development_vpc.id
}

output "dev-subnet-id" {
  value = aws_subnet.dev_subnet_1.id
}

output "dev-vpc-arn" {
  value = aws_vpc.development_vpc.arn
}

output "dev-subnet-arn" {
  value = aws_subnet.dev_subnet_1.arn
}

```

```

@manahilhabib031 → /workspaces/Lab9 (main) $ vim main.tf
"main.tf" 72L, 1644B writtenes/Lab9 (main) $

```

```

main.tf 72L 1044B writtenes/Lab9 (main) $
@manahilhabib031 → /workspaces/Lab9 (main) $ terraform apply
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-08964a93a7cb32cac]
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-021a5e129ca204b3f]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-098e7c5175da07629]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0440c0cbe438eeb57]

Changes to Outputs:
+ dev-subnet-arn = "arn:aws:ec2:me-central-1:095032391730:subnet/subnet-0440c0cbe438eeb57"
+ dev-subnet-id  = "subnet-0440c0cbe438eeb57"
+ dev-vpc-arn   = "arn:aws:ec2:me-central-1:095032391730:vpc/vpc-08964a93a7cb32cac"
+ dev-vpc-id    = "vpc-08964a93a7cb32cac"

You can apply this plan to save these new output values to the Terraform state, without changing any real
infrastructure.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

dev-subnet-arn = "arn:aws:ec2:me-central-1:095032391730:subnet/subnet-0440c0cbe438eeb57"
dev-subnet-id  = "subnet-0440c0cbe438eeb57"
dev-vpc-arn   = "arn:aws:ec2:me-central-1:095032391730:vpc/vpc-08964a93a7cb32cac"
dev-vpc-id    = "vpc-08964a93a7cb32cac"
@manahilhabib031 → /workspaces/Lab9 (main) $ |

```

Return ALL of the following output attributes from your VPC and Subnet resources:

- a) cidr_block
- b) region
- c) tags.Name
- d) tags_all

Expand output section in main.tf accordingly:

```

output "dev-vpc-cidr_block" {
  value = aws_vpc.development_vpc.cidr_block
}

output "dev-vpc-region" {
  value = aws_vpc.development_vpc.region
}

output "dev-vpc-tags_name" {
  value = aws_vpc.development_vpc.tags["Name"]
}

```



```
output "dev-vpc-tags_all" {  
    value = aws_vpc.development_vpc.tags_all  
}  
  
output "dev-subnet-cidr_block" {  
    value = aws_subnet.dev_subnet_1.cidr_block  
}  
  
output "dev-subnet-region" {  
    value = aws_subnet.dev_subnet_1.availability_zone  
}  
  
output "dev-subnet-tags_name" {  
    value = aws_subnet.dev_subnet_1.tags["Name"]  
}  
  
output "dev-subnet-tags_all" {  
    value = aws_subnet.dev_subnet_1.tags_all  
}
```

CA Command Prompt - gh codespace ssh

```
}  
  
output "dev-vpc-arn" {  
  value = aws_vpc.development_vpc.arn  
}  
  
output "dev-subnet-arn" {  
  value = aws_subnet.dev_subnet_1.arn  
}  
  
output "dev-vpc-cidr_block" {  
  value = aws_vpc.development_vpc.cidr_block  
}  
  
output "dev-vpc-region" {  
  value = aws_vpc.development_vpc.region  
}  
  
output "dev-vpc-tags_name" {  
  value = aws_vpc.development_vpc.tags["Name"]  
}  
  
output "dev-vpc-tags_all" {  
  value = aws_vpc.development_vpc.tags_all  
}  
  
output "dev-subnet-cidr_block" {  
  value = aws_subnet.dev_subnet_1.cidr_block  
}  
  
output "dev-subnet-region" {  
  value = aws_subnet.dev_subnet_1.availability_zone  
}  
  
output "dev-subnet-tags_name" {  
  value = aws_subnet.dev_subnet_1.tags["Name"]  
}  
  
output "dev-subnet-tags_all" {  
  value = aws_subnet.dev_subnet_1.tags_all  
}
```

Run terraform apply and record outputs.

```

@manahilhabib031 → /workspaces/Lab9 (main) $ terraform apply
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-08964a93a7cb32cac]
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-021a5e129ca204b3f]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-098e7c5175da07629]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0440c0cbe438eeb57]

Changes to Outputs:
+ dev-subnet-cidr_block = "10.0.10.0/24"
+ dev-subnet-region     = "me-central-1a"
+ dev-subnet-tags_all   = {
  + Name = "subnet-1-dev"
}
+ dev-subnet-tags_name  = "subnet-1-dev"
+ dev-vpc-cidr_block    = "10.0.0.0/16"
+ dev-vpc-region        = "me-central-1"
+ dev-vpc-tags_all      = {
  + Name = "development"
}
+ dev-vpc-tags_name     = "development"

You can apply this plan to save these new output values to the Terraform state, without changing
infrastructure.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

dev-subnet-arn = "arn:aws:ec2:me-central-1:095032391730:subnet/subnet-0440c0cbe438eeb57"
dev-subnet-cidr_block = "10.0.10.0/24"
dev-subnet-id = "subnet-0440c0cbe438eeb57"
dev-subnet-region = "me-central-1a"
dev-subnet-tags_all = tomap({
  "Name" = "subnet-1-dev"
})
dev-subnet-tags_name = "subnet-1-dev"
dev-vpc-arn = "arn:aws:ec2:me-central-1:095032391730:vpc/vpc-08964a93a7cb32cac"
dev-vpc-cidr_block = "10.0.0.0/16"
dev-vpc-id = "vpc-08964a93a7cb32cac"
dev-vpc-region = "me-central-1"
dev-vpc-tags_all = tomap({
  "Name" = "development"
})
dev-vpc-tags_name = "development"
@manahilhabib031 → /workspaces/Lab9 (main) $ |

```

Cleanup — Delete Resources & State Verification

Destroy all resources:

terraform destroy

```
@manahilhabib031 → /workspaces/Lab9 (main) $ terraform destroy
data.aws_vpc.existing_vpc: Reading...
aws_vpc.development_vpc: Refreshing state... [id=vpc-08964a93a7cb32cac]
data.aws_vpc.existing_vpc: Read complete after 1s [id=vpc-021a5e129ca204b3f]
aws_subnet.dev_subnet_1_existing: Refreshing state... [id=subnet-098e7c5175da07629]
aws_subnet.dev_subnet_1: Refreshing state... [id=subnet-0440c0cbe438eeb57]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  - destroy

Terraform will perform the following actions:

  # aws_subnet.dev_subnet_1 will be destroyed
  - resource "aws_subnet" "dev_subnet_1" {
    - arn                                = "arn:aws:ec2:me-central-1:095032391730:subnet/subnet-0440c0cbe438eeb57" -> null
    - assign_ipv6_address_on_creation    = false -> null
    - availability_zone                  = "me-central-1a" -> null
    - availability_zone_id                = "mec1-az1" -> null
    - cidr_block                         = "10.0.10.0/24" -> null
    - enable_dns64                       = false -> null
    - enable_lni_at_device_index         = 0 -> null
    - enable_resource_name_dns_a_record_on_launch = false -> null
    - enable_resource_name_dns_aaaa_record_on_launch = false -> null
    - id                                = "subnet-0440c0cbe438eeb57" -> null
    - ipv6_native                       = false -> null
    - map_customer_owned_ip_on_launch    = false -> null
    - map_public_ip_on_launch           = false -> null
    - owner_id                          = "095032391730" -> null
    - private_dns_hostname_type_on_launch = "ip-name" -> null
    - region                            = "me-central-1" -> null
    - tags                              = {
      - "Name" = "subnet-1-dev"
    } -> null
    - tags_all                          = {
      - "Name" = "subnet-1-dev"
    } -> null
    - vpc_id                            = "vpc-08964a93a7cb32cac" -> null
    # (4 unchanged attributes hidden)
  }

  # aws_subnet.dev_subnet_1_existing will be destroyed
  - resource "aws_subnet" "dev_subnet_1_existing" {
    - arn                                = "arn:aws:ec2:me-central-1:095032391730:subnet/subnet-098e7c5175da07629" -> null
    - assign_ipv6_address_on_creation    = false -> null
    - availability_zone                  = "me-central-1a" -> null
    - availability_zone_id                = "mec1-az1" -> null
    - cidr_block                         = "172.31.48.0/24" -> null
    - enable_dns64                       = false -> null
    - enable_lni_at_device_index         = 0 -> null
    - enable_resource_name_dns_a_record_on_launch = false -> null
    - enable_resource_name_dns_aaaa_record_on_launch = false -> null
    - id                                = "subnet-098e7c5175da07629" -> null
    - ipv6_native                       = false -> null
```

```
Plan: 0 to add, 0 to change, 3 to destroy.
```

```
Changes to Outputs:
```

```
- dev-subnet-arn      = "arn:aws:ec2:me-central-1:784386063361:subnet/subnet-08d3345f7fcf2eb31" -> null
- dev-subnet-cidr_block = "10.0.10.0/24" -> null
- dev-subnet-id       = "subnet-08d3345f7fcf2eb31" -> null
- dev-subnet-region    = "me-central-1a" -> null
- dev-subnet-tags_all  = {
  - Name = "subnet-1-dev"
} -> null
- dev-subnet-tags_name = "subnet-1-dev" -> null
- dev-vpc-arn          = "arn:aws:ec2:me-central-1:784386063361:vpc/vpc-0ff8ba45dd1117b74" -> null
- dev-vpc-cidr_block   = "10.0.0.0/16" -> null
- dev-vpc-id           = "vpc-0ff8ba45dd1117b74" -> null
- dev-vpc-region       = "me-central-1" -> null
- dev-vpc-tags_all     = {
  - Name = "development"
} -> null
- dev-vpc-tags_name    = "development" -> null
```

```
Do you really want to destroy all resources?
```

```
Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.
```

```
Enter a value: yes
```

```
aws_subnet.dev_subnet_1_existing: Destroying... [id=subnet-06e543931482a05cc]
aws_subnet.dev_subnet_1: Destroying... [id=subnet-08d3345f7fcf2eb31]
aws_subnet.dev_subnet_1_existing: Destruction complete after 1s
aws_subnet.dev_subnet_1: Destruction complete after 1s
aws_vpc.development_vpc: Destroying... [id=vpc-0ff8ba45dd1117b74]
aws_vpc.development_vpc: Destruction complete after 1s
```

```
Destroy complete! Resources: 3 destroyed.
```

Inspect state files:

```
cat terraform.tfstate
```

```
cat terraform.tfstate.backup
```

```

@manahilhabib031 → /workspaces/Lab9 (main) $ cat terraform.tfstate
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 42,
  "lineage": "8b1d5d5f-e9bc-aaf3-332c-e079db2b49e2",
  "outputs": {},
  "resources": [],
  "check_results": null
}
@manahilhabib031 → /workspaces/Lab9 (main) $ cat terraform.tfstate.backup
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 37,
  "lineage": "8b1d5d5f-e9bc-aaf3-332c-e079db2b49e2",
  "outputs": {
    "dev-subnet-arn": {
      "value": "arn:aws:ec2:me-central-1:095032391730:subnet/subnet-0440c0cbe438eeb57",
      "type": "string"
    },
    "dev-subnet-cidr_block": {
      "value": "10.0.10.0/24",
      "type": "string"
    },
    "dev-subnet-id": {
      "value": "subnet-0440c0cbe438eeb57",
      "type": "string"
    },
    "dev-subnet-region": {
      "value": "me-central-1a",
      "type": "string"
    },
    "dev-subnet-tags_all": {
      "value": {
        "Name": "subnet-1-dev"
      },
      "type": [
        "map",
        "string"
      ]
    },
    "dev-subnet-tags_name": {

```

Reapply then compare state and backup files to see differences.

```
@manahilhabib031 → /workspaces/Lab9 (main) $ terraform apply
```

```
data.aws_vpc.existing_vpc: Reading...
```

```
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-021a5e129ca204b3f]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# aws_subnet.dev_subnet_1 will be created
```

```
+ resource "aws_subnet" "dev_subnet_1" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "10.0.10.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch            = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
  + tags                               = {
    + "Name" = "subnet-1-dev"
  }
  + tags_all                           = {
    + "Name" = "subnet-1-dev"
  }
  + vpc_id                             = (known after apply)
}
```

```
# aws_subnet.dev_subnet_1_existing will be created
```

```
+ resource "aws_subnet" "dev_subnet_1_existing" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "172.31.48.0/24"
```

```
+ dev-vpc-arn          = (known after apply)
+ dev-vpc-cidr_block   = "10.0.0.0/16"
+ dev-vpc-id           = (known after apply)
+ dev-vpc-region       = "me-central-1"
+ dev-vpc-tags_all     = {
  + Name = "development"
}
+ dev-vpc-tags_name    = "development"
```

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

aws_subnet.dev_subnet_1_existing: Creating...

aws_vpc.development_vpc: Creating...

aws_subnet.dev_subnet_1_existing: Creation complete after 1s [id=subnet-07bb2...

aws_vpc.development_vpc: Creation complete after 2s [id=vpc-04e43bb38134534dd...

aws_subnet.dev_subnet_1: Creating...

aws_subnet.dev_subnet_1: Creation complete after 0s [id=subnet-0d9ad0b824814936c...

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

dev-subnet-arn = "arn:aws:ec2:me-central-1:784386063361:subnet/subnet-0d9ad0b824814936c"

dev-subnet-cidr_block = "10.0.10.0/24"

dev-subnet-id = "subnet-0d9ad0b824814936c"

dev-subnet-region = "me-central-1a"

dev-subnet-tags_all = tomap({

"Name" = "subnet-1-dev"

})

dev-subnet-tags_name = "subnet-1-dev"

dev-vpc-arn = "arn:aws:ec2:me-central-1:784386063361:vpc/vpc-04e43bb38134534dd"

dev-vpc-cidr_block = "10.0.0.0/16"

dev-vpc-id = "vpc-04e43bb38134534dd"

dev-vpc-region = "me-central-1"

dev-vpc-tags_all = tomap({

"Name" = "development"

})

dev-vpc-tags_name = "development"


```
@manahilhabib031 → /workspaces/Lab9 (main) $ cat terraform.tfstate
```

```
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 46,
  "lineage": "8b1d5d5f-e9bc-aaf3-332c-e079db2b49e2",
  "outputs": {
    "dev-subnet-arn": {
      "value": "arn:aws:ec2:me-central-1:095032391730:subnet/subnet-0e1b5658aaff50ae2",
      "type": "string"
    },
    "dev-subnet-cidr_block": {
      "value": "10.0.10.0/24",
      "type": "string"
    },
    "dev-subnet-id": {
      "value": "subnet-0e1b5658aaff50ae2",
      "type": "string"
    },
    "dev-subnet-region": {
      "value": "me-central-1a",
      "type": "string"
    },
    "dev-subnet-tags_all": {
      "value": {
        "Name": "subnet-1-dev"
      },
      "type": [
        "map",
        "string"
      ]
    },
    "dev-subnet-tags_name": {
      "value": "subnet-1-dev",
      "type": "string"
    },
    "dev-vpc-arn": {
      "value": "arn:aws:ec2:me-central-1:095032391730:vpc/vpc-0e3f9188fdd3bdc8b",
      "type": "string"
    },
    "dev-vpc-cidr_block": {
      "value": "10.0.0.0/16",
      "type": "string"
    },
    "dev-vpc-id": {
```

```
@manahilhabib031 → /workspaces/Lab9 (main) $ cat terraform.tfstate.backup
```

```
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 42,
  "lineage": "8b1d5d5f-e9bc-aaf3-332c-e079db2b49e2",
  "outputs": {},
  "resources": [],
  "check_results": null
}
```

```
@manahilhabib031 → /workspaces/Lab9 (main) $ |
```